PORTFOLIO PROJECT

Privacy-Aware AI Gmail Message Classifier

A production-oriented inbox workflow that classifies individual unread messages with OpenAI while preserving user state, minimizing transmitted data, and keeping failures reviewable.

PLATFORM	AI CONTRACT	PRIVACY	QUALITY
Google Apps Script	**Strict Structured Outputs**	**Minimized and redacted input**	**30 tests + 94% branches**

Business challenge

Business inboxes mix invoices, orders, complaints, quote requests, promotions, and internal communication. Manual triage is repetitive, while careless automation can skip new replies, duplicate work, expose sensitive content, or create unbounded provider cost.

Implemented workflow

The classifier processes each eligible unread message, builds a minimized and redacted input, requests a schema-constrained category from the OpenAI Responses API, validates the result locally, and applies namespaced Gmail labels without marking the message as read.

Message-level data flow

1. SELECT	2. MINIMIZE	3. CLASSIFY	4. APPLY
Query unread inbox messages and exclude managed completion or review labels.	Use sender domain, redacted subject, and Gmail snippet. Skip bodies and attachments.	Responses API with store: false, strict schema, local enums, and confidence threshold.	Add AI category and AI/Processed atomically; route permanent failures to AI/Needs Review.

Why message-level processing matters

A newly received reply remains eligible even when an earlier message in the same conversation was already classified. This avoids a common thread-level automation bug and preserves the user's unread workflow.

RELIABILITY, PRIVACY, OPERATIONS

Designed for controlled automation

The implementation combines bounded provider behavior with explicit Gmail ownership rules and operational recovery paths.

Reliability controls	Privacy and security
- LockService prevents overlapping trigger runs	- API key stays in Script Properties
- Run deadline leaves remaining work for the next trigger	- Best-effort PII redaction before provider call
- Bounded exponential backoff for transient failures	- Metadata-only, formula-safe error logs
- Idempotent setup and trigger creation	- Explicit OAuth scope allowlist

Operational visibility

- AI/Processed records successful completion without changing unrelated labels.
- AI/Needs Review creates a visible queue for permanent message-level failures.
- Safe status helpers expose configuration readiness without returning secrets.
- Client and provider request IDs support correlation without logging message content.
- Configurable 90-day default retention automatically removes expired error rows.

Verification evidence

30 deterministic tests cover minimized input, PII redaction, strict output validation, full message orchestration, retry and error branches, config invariants, idempotent setup, error-sheet recovery, retention, label updates, and spreadsheet safety. Instrumented coverage is 96.95% statements, 94.03% branches, 100% functions, and 96.89% lines.

Evidence status

Verified locally on 2026-07-17 at code commit 607039a9e53a051694e7348cdd33cc2419bb7773. Live Apps Script verification: pending. Portfolio verification only; no client deployment or measured business outcome is claimed. Repository: github.com/bazhinapolly/ai-gmail-classifier-apps-script

Deployment path

Install with clasp or the Apps Script editor, add a restricted OpenAI project key in Script Properties, review the manifest scopes, run idempotent setup, execute the controlled smoke test, and validate with a dedicated Gmail account before enabling the five-minute trigger.

Business value

The workflow is designed to reduce repetitive inbox triage, keep ambiguous cases visible, preserve the user's normal email state, and provide a controlled foundation for account-specific categories and policies.